

2016.0 RANGE ROVER (LG), 206-11

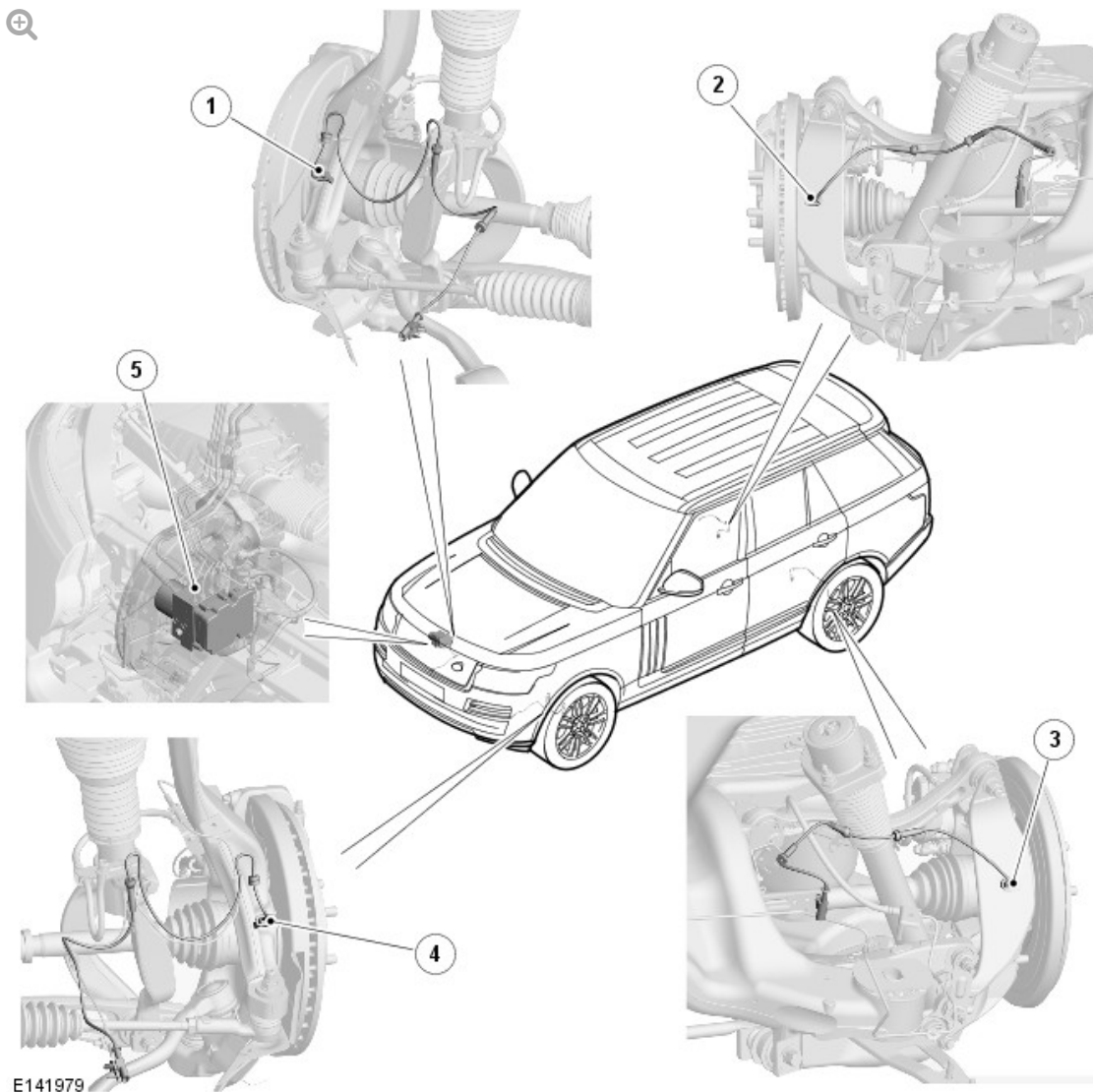
BRAKE CONTROLS

DESCRIPTION AND OPERATION

COMPONENT LOCATION - SHEET 1 OF 2

NOTE:

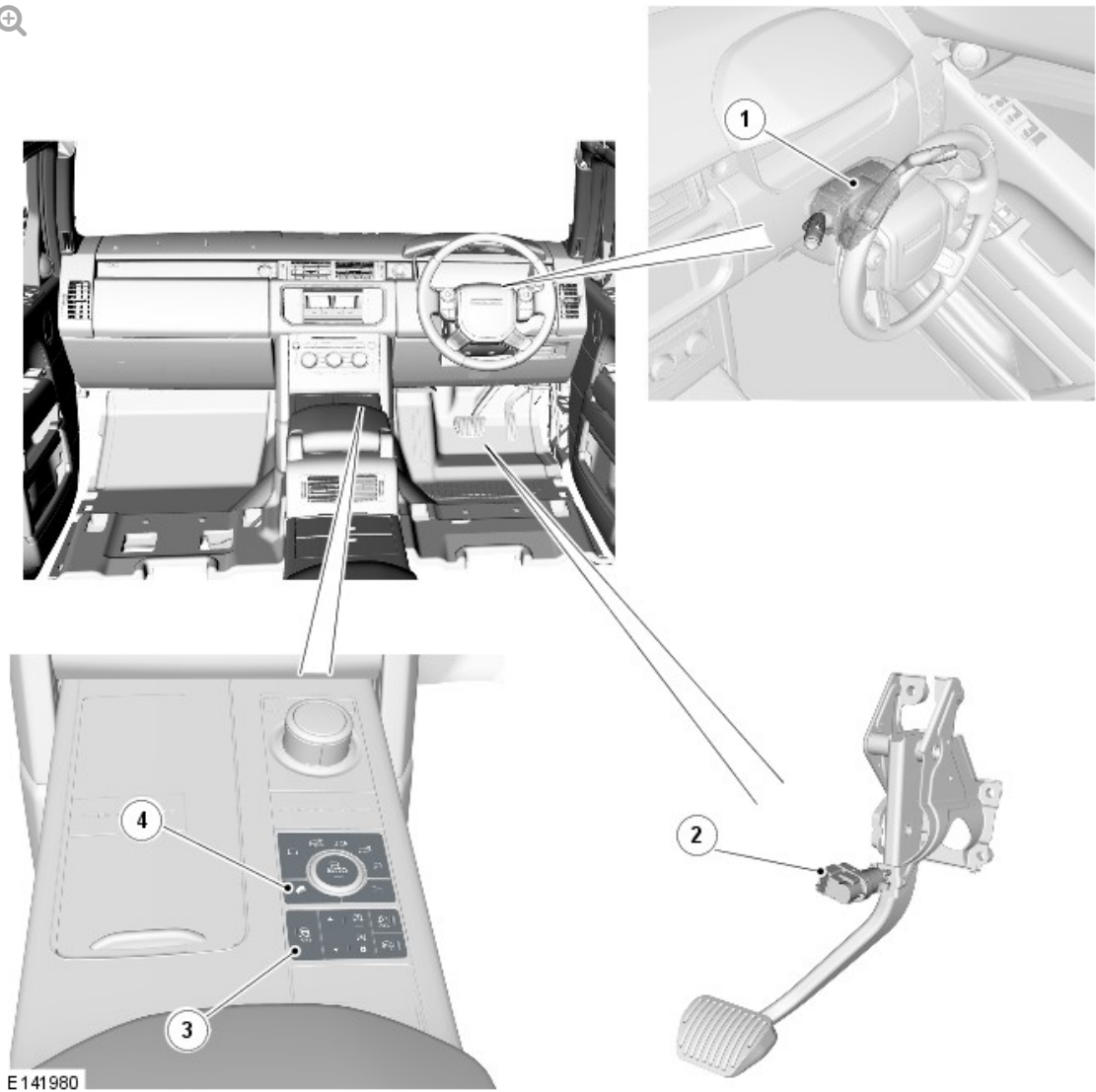
RHD (right-hand drive) installations shown, LHD (left-hand drive) installations similar.



ITEM	DESCRIPTION
1	Right front wheel speed sensor
2	Right rear wheel speed sensor
3	Left rear wheel speed sensor
4	Left front wheel speed sensor
5	Integrated anti-lock brake system control module and hydraulic control unit

NOTE:

RHD installations shown, LHD installations similar.



ITEM	DESCRIPTION
1	Steering angle sensor (integral with clockspring)
2	Brake pedal switch
3	Dynamic stability control switch
4	Hill descent control switch

OVERVIEW

The braking control system features an integrated four-channel HCU (hydraulic control unit) and ABS (anti-lock brake system) control module.

The ABS control module is connected to the high speed CAN (controller area network) powertrain systems and the high speed CAN chassis systems, and actively interacts with other vehicle system control modules and associated sensors to receive and transmit current vehicle operating information.

When required, the ABS control module will actively intervene and operate the HCU during braking or vehicle maneuvers to correct the vehicle attitude, stability, traction or speed. During incidents of vehicle correction, the ABS control module may also request the ECM (engine control module) to control engine power in order to further stabilize and correct the vehicle.

To provide full system functionality, the braking control system comprises the following components:

- DSC (dynamic stability control) switch.
- HDC (hill descent control) switch.
- Four wheel speed sensors.
- Steering angle sensor.
- Brake pedal switch.
- Instrument cluster warning indicators.
- ABS control module with integrated HCU.

NOTE: Unlike some previous brake control systems, this system does not have a separate lateral acceleration and yaw rate sensor component to provide vehicle acceleration data. Instead, the ABS control module receives vehicle acceleration data over the high speed CAN powertrain systems from the RCM (restraints control module).

For additional information, refer to: [Airbag and Safety Belt Pretensioner](#)

[Supplemental Restraint System](#) (501-20B Supplemental Restraint System, Description and Operation).

The braking control system provides the following brake functions that are designed to assist the vehicle or aid the driver:

- ABS.
- RSC (roll stability control).
- CBC (corner brake control).
- DSC.
- EBD (electronic brake force distribution).
- Electronic brake prefill.
- ETC (electronic traction control).
- EBA (emergency brake assist).
- EDC (engine drag-torque control).
- EUC (enhanced understeer control).
- Gradient acceleration control.
- Gradient release control.
- HDC.
- Hill start assist.
- Terrain Response® system integration.
- Trailer stability assist.

The HDC function is enabled either manually using the HDC switch, or automatically by the Terrain Response® system, in the ignition on and engine running power modes. All of the other brake functions are automatically enabled in the ignition on and engine running power modes. The DSC function can be selected off using the DSC switch.

DESCRIPTION

DYNAMIC STABILITY CONTROL SWITCH



E141981

The DSC switch allows the DSC function to be selected off. Although Land Rover recommend that DSC is selected on for all normal driving conditions, it may be beneficial to de-select DSC to maximize traction under the following conditions:

- If the vehicle needs to be rocked out of a hollow or a soft surface.
- Driving on loose surfaces or with snow chains.
- Driving in deep sand, snow or mud.
- On tracks with deep longitudinal ruts.

The DSC switch is a non-latching switch installed in the Terrain Response® switchpack, in the floor console. Pressing the DSC switch produces a high speed CAN chassis systems message from the Terrain Response® switchpack to the ABS control module. With the first press of the DSC switch, the ABS control module disables the DSC functions. When the DSC switch is pressed again, the ABS control module re-enables the DSC functions. The DSC switch must be pressed for a minimum of 0.3 s for the ABS control module to react. The DSC function is re-enabled at the beginning of each ignition cycle.

The status of the DSC switch selection is shown by the DSC OFF warning indicator. The DSC OFF warning indicator is extinguished while DSC is selected on, and continuously illuminated while DSC is selected off.

A DSC switch request to disable DSC is ignored by the ABS control module if the air suspension system has failed, or is in off-road height at speeds above 55 km/h (34 mph).

To guard against incorrect operation or a broken switch, if the input from the DSC switch is held high for more than one minute, a DTC (diagnostic trouble code) is stored in the ABS control module.

Even if DSC is deselected, driving maneuvers with extreme yaw or lateral acceleration may trigger DSC activity to assist vehicle stability.

HILL DESCENT CONTROL SWITCH



E141982

The HDC switch controls the selection of the HDC function.

The HDC switch is a non-latching switch installed in the Terrain Response® switchpack, in the floor console. Pressing and releasing the HDC switch produces a high speed CAN chassis systems message from the Terrain Response® switchpack to the ABS control module. With the first press and release of the HDC switch, the ABS control module enables operation of the HDC function. When the HDC switch is pressed and released again, the ABS control module disables operation of the HDC function.

To guard against incorrect operation or a broken switch, if the switch is pressed for more than 10 seconds no change of state occurs. If the input from the HDC switch is held high for more than one minute, a DTC is stored in the ABS control module.

WHEEL SPEED SENSORS



E141983

An active wheel speed sensor is installed in each wheel hub to provide the ABS control module with a rotational speed signal from each road wheel. The head of each wheel speed sensor is positioned close to a magnetic encoder incorporated into the seal of the wheel bearing. A flying lead connects each sensor to the vehicle wiring.

The wheel speed sensors each have a power supply connection and a signal connection with the ABS control module. When the ignition is on, the ABS control module supplies power to the wheel speed sensors and monitors the return signals. The return signals are converted into individual wheel speeds and the overall vehicle speed by the ABS control module.

The ABS control module outputs the individual wheel speeds, the vehicle speed and the direction of travel on the high speed CAN buses for use by other systems. The quality of the vehicle speed signal is also broadcast on the high speed CAN buses. If all wheel speed signals are available to calculate vehicle speed from, the quality of the vehicle speed signal is set to 'data calculated within specified accuracy'. If one or more wheel speed sensors is faulty, the quality of the vehicle speed signal is set to 'accuracy

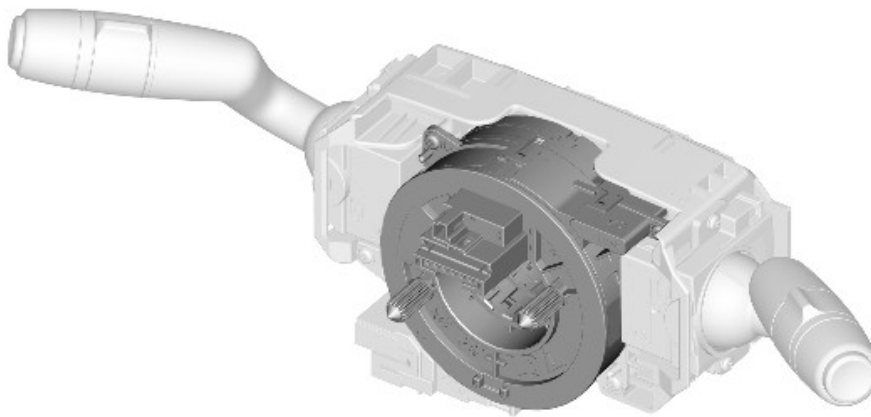
outside specification'.

The ABS control module monitors the wheel speed sensor circuits for faults. If a fault is detected the ABS control module stores a DTC and illuminates the appropriate warning indicators, depending on the system functions affected (DSC/ETC, ABS, EBA/EBD, HDC). A warning chime is also sounded to alert the driver to the fault condition, and a related message is shown in the message center.

For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument Cluster, Description and Operation).

Since the wheel speed sensors are active devices, a return signal is available when the road wheels are not turning, which enables the ABS control module to check the sensors while the vehicle is stationary. In addition, the direction of travel of each wheel can be sensed. This information is broadcast on the high speed CAN buses for use by other systems.

STEERING ANGLE SENSOR



E141984

The steering angle sensor is integrated into the clockspring on the steering column and is able to measure steering wheel rotation from lock to lock.

Steering angle sensor signals are processed in the clockspring to calculate the steering wheel angle and the steering wheel angle speed. These values, together with signal integrity information, is transmitted on the high speed CAN chassis systems for use by the ABS control module and functions such as DSC.

If a fault occurs within the steering angle sensor, a DTC is stored in the steering angle sensor and the ABS control module. The ABS control module signals the instrument cluster on the high speed CAN powertrain systems to illuminate the appropriate warning indicators, depending on the brake functions affected. A warning chime sounds to alert the driver to the fault condition and, if the fault affects the HDC function, a message is displayed in the message center.

BRAKE PEDAL SWITCH



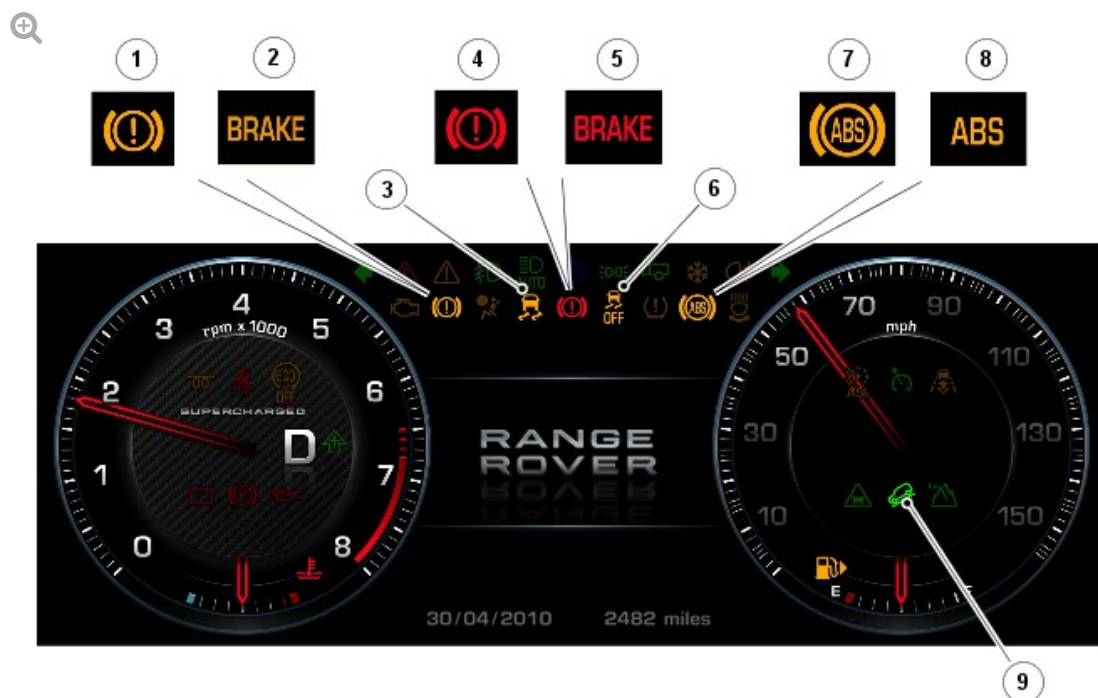
E125620

The brake pedal switch is a two pole switch mounted in the brake pedal bracket and operated by the brake pedal. The ECM compares the signals from the two poles to confirm the status of the brake pedal switch, which it broadcasts on the high speed CAN powertrain systems for the ABS control module.

NOTE:

The output from both poles is also connected to the CJB (central junction box)

INSTRUMENT CLUSTER WARNING INDICATORS

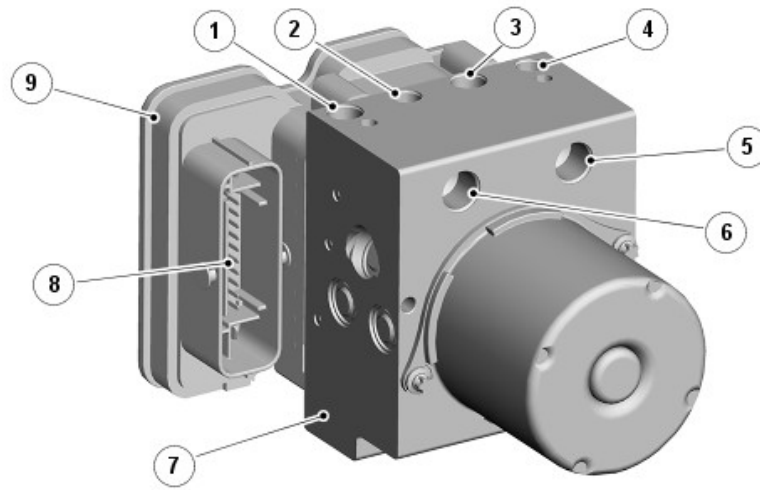


E141985

ITEM	DESCRIPTION
1	Brake amber warning indicator (all except Canada, Mexico and USA)
2	Brake amber warning indicator (Canada, Mexico and USA)
3	DSC warning indicator
4	Brake red warning indicator (Canada, Mexico and USA)
5	Brake red warning indicator (all except Canada, Mexico and USA)
6	DSC OFF warning indicator
7	ABS warning indicator (all except Canada, Mexico and USA)
8	ABS warning indicator (Canada, Mexico and USA)
9	HDC warning indicator

The instrument cluster contains various warning indicators for the braking control functions. The warning indicators provide a visual indication of either a system fault or system operating status. Except for HDC, the warning indicators illuminate briefly when the ignition is switched on, as a 'bulb' check.

ANTI-LOCK BRAKE SYSTEM CONTROL MODULE



E142693

ITEM	DESCRIPTION
1	Rear right brake connection
2	Rear left brake connection
3	Front right brake connection
4	Front left brake connection
5	Primary circuit connection
6	Secondary circuit connection
7	Hydraulic control unit
8	Electrical connector
9	Control module

The ABS control module consists of an integrated control module and four-channel HCU. The control module uses the HCU to modulate hydraulic pressure to the individual wheel brakes to control the brake functions.

The ABS control module is installed in the front corner of the engine compartment, on the driver side. A multi-pin electrical connector provides the electrical interface between the control module and the vehicle wiring. Hydraulic pipes and hoses connect the HCU to the master cylinder and the brake calipers. The primary and secondary outlets of the master cylinder are connected to the primary and secondary circuits within the HCU. The

primary circuit in the HCU has separate outlet ports to the front brakes. The secondary circuit in the HCU has separate outlet ports to the rear brakes.

The ABS control module operates in three modes: Normal braking/EBD, ABS braking and active braking.

NORMAL BRAKING/ELECTRONIC BRAKE FORCE DISTRIBUTION MODE

Operating the brake pedal produces a corresponding increase or decrease of pressure in the brakes, unaffected by the ABS control module. If the ABS control module determines that EBD is necessary, it isolates the brakes of the trailing axle from any further increase in hydraulic pressure.

ABS BRAKING MODE

If the ABS control module determines that ABS braking is necessary, it releases pressure from the affected brake. The reduced pressure allows the wheel to accelerate. The ABS control module then regulates the pressure in the brake to apply the maximum braking effort without locking the wheel. Individual ABS braking is provided for each wheel.

ACTIVE BRAKING MODE

In the active braking mode, the ABS control module generates and controls hydraulic pressure to the brakes for functions other than ABS braking, for example DSC, EBA, ETC, HDC, and in support of adaptive speed control and the dynamic application of the EPB (electronic parking brake).

The ABS control module regulates the pressure in the individual brakes as necessary. Some noise may be generated during active braking.

OPERATION

ANTI-LOCK BRAKE SYSTEM

ABS controls the speed of all road wheels to ensure optimum wheel slip when braking at the adhesion limit. This prevents the wheels from locking in

order to retain effective steering control of the vehicle.

On each axle, the brake pressure is modulated separately for each wheel in order to maintain vehicle stability.

ROLL STABILITY CONTROL

The RSC function uses the brake system and engine torque control to attempt to restore vehicle stability if the vehicle is forced into such a harsh manoeuvre that it risks tipping over.

The ABS control module monitors driver inputs and vehicle behavior using various powertrain signals and inputs from the wheel speed sensors, steering angle sensor and the RCM (yaw rate and lateral acceleration data). These are compared with modeled behavior and, if vehicle behavior reaches a given risk level, the ABS control module initiates a reduction in engine power, or brakes one or more wheels sufficiently to correct the vehicle stability and assist the driver to remain in control.

While the ignition is on, RSC is permanently enabled, even when DSC is selected off.

CORNER BRAKE CONTROL

CBC influences the brake pressures, below the DSC and ABS thresholds, to counteract the yawing moment produced when braking in a corner. CBC produces a correction torque by limiting the brake pressure on one side of the vehicle, to assist the vehicle in achieving the turn radius the driver is requesting.

DYNAMIC STABILITY CONTROL

DSC uses the brakes and engine torque control to help maintain the lateral stability of the vehicle. While the ignition is on the DSC function is permanently enabled unless selected off by the DSC switch. Even if DSC is deselected, driving maneuvers with extreme yaw or lateral acceleration may trigger DSC activity to assist vehicle stability.

DSC enhances driving safety in abrupt maneuvers and in understeer or

oversteer situations that may occur in a bend. The ABS control module monitors the yaw rate and lateral acceleration of the vehicle, and the steering input, then selectively applies individual brakes and signals for engine torque adjustments to reduce understeer or oversteer.

In general:

- In an understeering situation, the inner wheels are braked to counteract the yaw movement towards the outer edge of the bend.
- In an oversteering situation, the outer wheels are braked to prevent the rear end of the vehicle from pushing towards the outer edge of the bend.

The ABS control module monitors the tracking stability of the vehicle using inputs from the wheel speed sensors, the steering angle sensor and the RCM (yaw rate and lateral acceleration data). The tracking stability is compared with stored target data and, whenever the tracking stability deviates from the target data, the ABS control module intervenes by applying the appropriate brakes. When the DSC function is active, the ABS control module also signals the TCM (transmission control module) to prevent gear shifts, and the instrument cluster to flash the DSC warning indicator.

If necessary, the ABS control module also signals:

- The ECM, to reduce engine torque.
- The transfer box control module, to adjust the locking torque of the center differential.
- The rear differential control module, to adjust the locking torque of the rear differential.

The DSC function overrides the differential locking torque requests from the Terrain Response® system.

If DSC is selected off, a DSC system off message is displayed in the message center.

For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument Cluster, Description and Operation).

ELECTRONIC BRAKE FORCE DISTRIBUTION

EBD limits the brake pressure applied to the brakes of the trailing axle. When the brakes are applied, the weight transfer of the vehicle reduces the adhesion of the trailing axle wheels on the road surface. This may cause the wheels to slip and make the vehicle unstable.

EBD uses the ABS hardware to automatically optimize the pressure of the trailing axle brakes, below the point where ABS intervention is normally invoked. Only the trailing axle wheels are under EBD control.

ELECTRONIC BRAKE PREFILL

Electronic brake prefill senses any rapid throttle lift off, activating a small brake hydraulic pressure build-up of approximately 3 to 5 bar (43.5 to 72.5 lbf/in²) in anticipation of the brakes being applied. This gives a quicker brake pedal response and consequently slightly shorter stopping distances.

When the ABS control module detects rapid throttle lift off (from the signals received from the ECM over the high speed CAN powertrain systems), it applies a low brake pressure to all of the brakes to assist in a quicker brake application.

ELECTRONIC TRACTION CONTROL

ETC attempts to optimize forward traction by reducing engine torque or braking a spinning wheel until traction is restored.

ETC is activated if an individual wheel speed is above that of the vehicle reference speed (positive slip) and the brake pedal is not pressed. The spinning wheel is braked, allowing the excess torque to be transmitted to the non spinning wheels through the drive line. If necessary, the ABS control module also transmits a high speed CAN powertrain systems message to the ECM requesting a reduction in engine torque.

Torque reduction requests are for either a slow or fast response. A slow

response requests a reduction of throttle angle; a fast response requests an ignition cut-off.

When the DSC function is selected off with the DSC switch, the engine torque reduction feature is disabled.

When the ETC function is active the ABS control module also signals the TCM to prevent gear shifts.

EMERGENCY BRAKE ASSIST

EBA assists the driver during emergency braking situations by automatically maximizing the braking effort. There are two situations when the ABS control module will invoke EBA:

- When the brake pedal is pressed rapidly.
- When the brake pedal is pressed hard enough to bring the front brakes into ABS operation.

When the brake pedal is pressed rapidly, the ABS control module increases the hydraulic pressure to all of the brakes until the threshold for ABS operation is reached. This applies the maximum braking effort for the available traction. The ABS control module monitors for the rapid application of the brakes using the brake pedal switch status broadcast on the high speed CAN powertrain systems by the ECM, and from a pressure sensor within the HCU. With the brake pedal pressed, if the rate of increase of hydraulic pressure exceeds the predetermined limit, the ABS control module invokes emergency braking.

When the brake pedal is pressed hard enough to bring the front brakes into ABS operation, the ABS control module increases the hydraulic pressure to the rear brakes up to the ABS threshold.

EBA operation continues until the driver releases the brake pedal enough for the hydraulic pressure in the HCU to fall below a pre-determined threshold value stored in the ABS control module.

ENGINE DRAG-TORQUE CONTROL

EDC prevents wheel slip caused by any of the following:

- A sudden decrease in engine torque when the accelerator is suddenly released.
- A downshift using CommandShift™.

When the ABS control module detects the onset of wheel slip without the brakes being applied, it signals the ECM on the high speed CAN powertrain systems to request a momentary increase in engine torque. Also, when the driver is braking and the ABS module has reduced brake pressure, but the wheels have not accelerated to the vehicle reference speed quickly enough, EDC will increase engine torque to accelerate the wheels up to reference speed and regain stability.

ENHANCED UNDERSTEER CONTROL

EUC monitors the vehicle for understeer by comparing signals of yaw rate and lateral acceleration, from the RCM, with signals from the steering angle sensor and wheel speed sensors.

When the ABS control module detects the onset of understeer, it signals the ECM via the high speed CAN powertrain systems to request a decrease in engine torque. If required, the ABS control module applies brake pressure to the inside rear wheel to correct the understeer. If the vehicle continues to understeer, EUC is activated and uses multiple brakes (maximum of three brakes) to rapidly reduce the vehicle speed.

GRADIENT ACCELERATION CONTROL

Gradient acceleration control is an automatic feature and is always available when HDC is not selected.

When HDC is not selected, gradient acceleration control will intervene to limit downhill acceleration on a steep descent.

The feature uses generated brake pressure to control acceleration in situations where the driver could lose control of the vehicle on a steep incline.

Gradient acceleration control keeps the vehicle to a speed and throttle pedal dependant acceleration limit when the vehicle is moving in the intended direction of travel, for example:

- Descending an incline forwards, with D (drive) selected.
- Descending an incline backwards, with R (reverse) selected.

When the vehicle is moving against the intended direction of travel, for example: descending a slope, but facing uphill with D selected, gradient acceleration control will prevent the vehicle accelerating above 5 km/h (3 mph) for up to 20-30 seconds to help the driver re-establish control of the vehicle.

GRADIENT RELEASE CONTROL

Gradient release control is an automatic feature which is always available when HDC is selected.

If the vehicle is brought to a standstill on a slope using the foot brake, gradient release control will become active (except in the Terrain Response®, sand program).

When descending a hill, a brake hold and gradual release is employed to provide a smooth transition into HDC. Gradient release control operates in forward and reverse gears and requires no driver intervention.

HILL DESCENT CONTROL

HDC uses brake intervention to control vehicle speed and acceleration during low speed descents in off-road and low grip on-road conditions. Generally, equal pressure is applied to all four brakes, but pressure to individual brakes can be modified by the ABS and DSC functions to retain stability. Selection of the HDC function is controlled by the HDC switch and the Terrain Response® switchpack in the floor console. HDC operates in both high and low ranges, at vehicle speeds up to 50 km/h (31.3 mph).

WARNING:

Incorrect use of the HDC function may compromise the stability of the vehicle, resulting in a dangerous and uncontrolled hill descent. Driving with the transmission in neutral while HDC is active will prevent engine braking from assisting the vehicle. The brakes will overheat and induce the HDC fade out strategy. In this condition there will be no control over the vehicle during a descent.

NOTE:

With the HDC function selected, HDC is operative even when the transmission is in neutral. It is not recommended to drive the vehicle further than is absolutely necessary with HDC selected and the transmission in neutral.

HDC may be used in D, R and all CommandShift™ gears. When in D, the TCM will automatically select the most appropriate gear.

HDC can be selected at speeds up to 80 km/h (50 mph), but will only be enabled at speeds below 50 km/h (31.3 mph). When HDC is selected:

- At speeds up to 50 km/h (31.3 mph), the HDC warning indicator is permanently illuminated if a valid gear is selected.
- At speeds from between 50 to 80 km/h (31 to 50 mph) the HDC warning indicator flashes and a message advising that the speed is too high is displayed in the message center. If the HDC switch is pressed while vehicle speed is more than 80 km/h (50 mph), the HDC warning indicator will not illuminate and HDC will not be selected.
- If the speed increases to 80 km/h (50 mph), the HDC function is switched off, the warning indicator is extinguished, a warning chime sounds and a message advising that HDC has been switched off is displayed in the

message center.

When HDC is enabled, the ABS control module calculates a target deceleration value by comparing the set speed to the actual vehicle speed. The ABS control module then operates the HCU in the active braking mode as required to achieve and maintain the target speed.

During active braking for HDC, the ABS control module sends a high speed CAN powertrain systems message to the CJB to operate the stop lamps. For additional information, refer to: [Exterior Lighting](#) (417-01 Exterior Lighting, Description and Operation).

Applying the foot brake during active braking may result in a pulse through the brake pedal.

The target speed varies between the minimum and maximum values for each gear and transmission range, depending on driver input with the speed control + and - switches. For additional information, refer to:

[Speed Control](#) (310-03A Speed Control - TDV6 3.0L Diesel, Description and Operation),

[Speed Control](#) (310-03B Speed Control - V6 S/C 3.0L Petrol, Description and Operation),

[Speed Control](#) (310-03B Speed Control - V6 S/C 3.0L Petrol, Description and Operation),

[Speed Control](#) (310-03B Speed Control - V6 S/C 3.0L Petrol, Description and Operation).

Low Range Target Speeds

TARGET SPEED	SPEED, KM/H (MPH)					
	R	D	1ST	2ND	3RD	4TH TO 8TH
Default	3.5 (2.2)	6 (3.7)*	6 (3.7)*	6 (3.7)*	8 (5.0)	10 (6.2)
Minimum	3.5 (2.2)	3.5 (2.2)	3.5 (2.2)	5 (3.1)	8 (5.0)	10 (6.2)
Maximum	25 (15.5)	25 (15.5)	25 (15.5)	25 (15.5)	25 (15.5)	25 (15.5)

* Default speed may be modified depending on Terrain Response® program and gear

High Range Target Speeds

TARGET SPEED	SPEED, KM/H (MPH)			
	R	D	1ST	2ND TO 8TH
Default	3.5 (2.2)	10 (6.2)*	10 (6.2)*	10 (6.2)*
Minimum	3.5 (2.2)	6 (3.7)	6 (3.7)	9 (5.6)
Maximum	35 (21.7)	35 (21.7)	35 (21.7)	35 (21.7)
* Default speed may be modified depending on Terrain Response® program and gear selection				

During changes of target speed, the ABS control module limits deceleration and acceleration to -0.5 m/s^2 (-1.65 ft/s^2) and $+0.5 \text{ m/s}^2$ ($+1.65 \text{ ft/s}^2$) respectively.

To provide a safe transition from active braking to brakes off, the ABS control module invokes a fade out strategy that gradually releases the braking effort during active braking. The fade out strategy occurs if any of the following conditions are detected during active braking:

- HDC selected off with the HDC switch.
- Failure of a component used by HDC, but not critical to the fade out function.
- Accelerator pedal pressed when the transmission is in neutral.
- Brake overheat.

If fade out is invoked because of deselection or component failure, the HDC function is cancelled by the ABS control module. If fade out is invoked because the accelerator pedal is pressed with the transmission in neutral, or because of brake overheat, the HDC function remains in standby and resumes operation when the accelerator pedal is released or the brakes have cooled.

The fade out strategy increases the target speed, at a constant acceleration rate of 0.5 m/s^2 (1.65 ft/s^2), until the maximum target speed is reached or until no active braking is required for 0.5 s. If the accelerator pedal is positioned within the range that influences target speed, the acceleration rate is increased to 1.0 m/s^2 (3.3 ft/s^2).

When fade out is invoked because of component failure, a warning chime sounds, the HDC warning indicator is extinguished and a message advising there is a fault is displayed in the message center.

When fade out is invoked because of brake overheat, a message advising that HDC is temporarily unavailable is displayed. At the end of fade out, the HDC warning indicator flashes. The message and flashing warning indicator remain on, while HDC remains selected, until the brakes have cooled.

To monitor for brake overheat, the ABS control module monitors the amount of braking activity and, from this, estimates the temperature of each brake. If the estimated temperature of any brake exceeds a preset limit, the ABS control module invokes the fade out strategy. After the fade out cycle, the HDC function is re-enabled when the ABS control module estimates that all of the brake temperatures are at less than 64% of the temperature limit.

When HDC is selected off, the instrument cluster message center temporarily displays a system off message.

ACTIVATION OF STOP LAMPS

Operation of the vehicle stop lamps is controlled by the CJB. The ABS control module monitors the brake system hydraulic pressure and requests the CJB, via the high speed CAN powertrain systems, to energize the stop lamps during active braking for HDC. A pressure threshold and time filter prevents the stop lamps from flickering.

STOP/START VEHICLES

Activation of HDC will deactivate the stop/start system. If HDC is activated while the engine is shutdown in a stop/start cycle, the engine will

automatically restart.

For additional information, refer to: [Starting System](#) (303-06A Starting System - TDV6 3.0L Diesel, Description and Operation).

HILL START ASSIST

Hill start assist is an automatic feature that aids smooth transition from foot brake to moving away on hill ascents, regardless of HDC selection.

On steep slopes, hill start assist will hold a portion of the driver generated brake pressure for a short time (2 to 3 seconds) to allow the driver to move their foot from the brake pedal to the throttle pedal without the vehicle rolling back.

The system will release the brake pressure in a controlled manner, either after the timer has expired or if the driver has generated sufficient drive-torque to move the vehicle forward up the hill.

The feature is not driver selectable and there is no indication to the driver when in operation.

TERRAIN RESPONSE® SYSTEM INTEGRATION

The Terrain Response® system integrates the ABS and other vehicle system control modules to assist the vehicle when driving off-road or during difficult surface conditions. Terrain Response® system integration is activated when a Terrain Response® special program is selected.

When a Terrain Response® special program is selected, the ABS control module, along with other vehicle system control modules, will operate in accordance with programmed software maps. The software maps allow the ABS system to function with a threshold that will assist the selected Terrain Response® special program.

For additional information, refer to: [Ride and Handling Optimization](#) (204-06 Ride and Handling Optimization, Description and Operation).

TRACTION LAUNCH CONTROL

Traction launch control is automatically enabled when either the sand or

grass/gravel/snow special program is selected on the Terrain Response® system.

When pulling away from stationary on sand or other dry, yielding ground, excessive wheel spin can cause the wheels to dig downwards preventing forward movement. On surfaces with low grip, excessive wheel spin can lead to vehicle instability and reduced vehicle acceleration.

Traction launch control uses the ETC function to limit the amount of wheel spin, allowing a gradual controlled pull away even if full throttle is applied until the vehicle has gained speed.

ROCK CRAWL PRECHARGE

Rock crawl precharge is automatically enabled when the rock crawl special program is selected on the Terrain Response® system.

Rock crawl precharge applies a small amount of brake pressure to each brake calliper during low speed driving. This improves brake and traction control response times helping to reduce forward/backward vehicle roll when cresting an obstacle or releasing the accelerator pedal.

TRAILER STABILITY ASSIST

When the trailer electrical socket is connected, trailer stability assist operates automatically to enhance the existing DSC and Terrain Response® functions of the vehicle when towing. The system detects sway movements caused by trailer oscillations at speeds in excess of 60 km/h (37 mph) and acts to eliminate them. It does this through braking and engine management. Braking management counterbalances the sway movement through symmetric and asymmetric braking, thereby slowing the vehicle and eliminating the oscillations. Engine management adapts engine torque output to support the braking management in stabilizing the vehicle and trailer.

Typical conditions when sway can occur include:

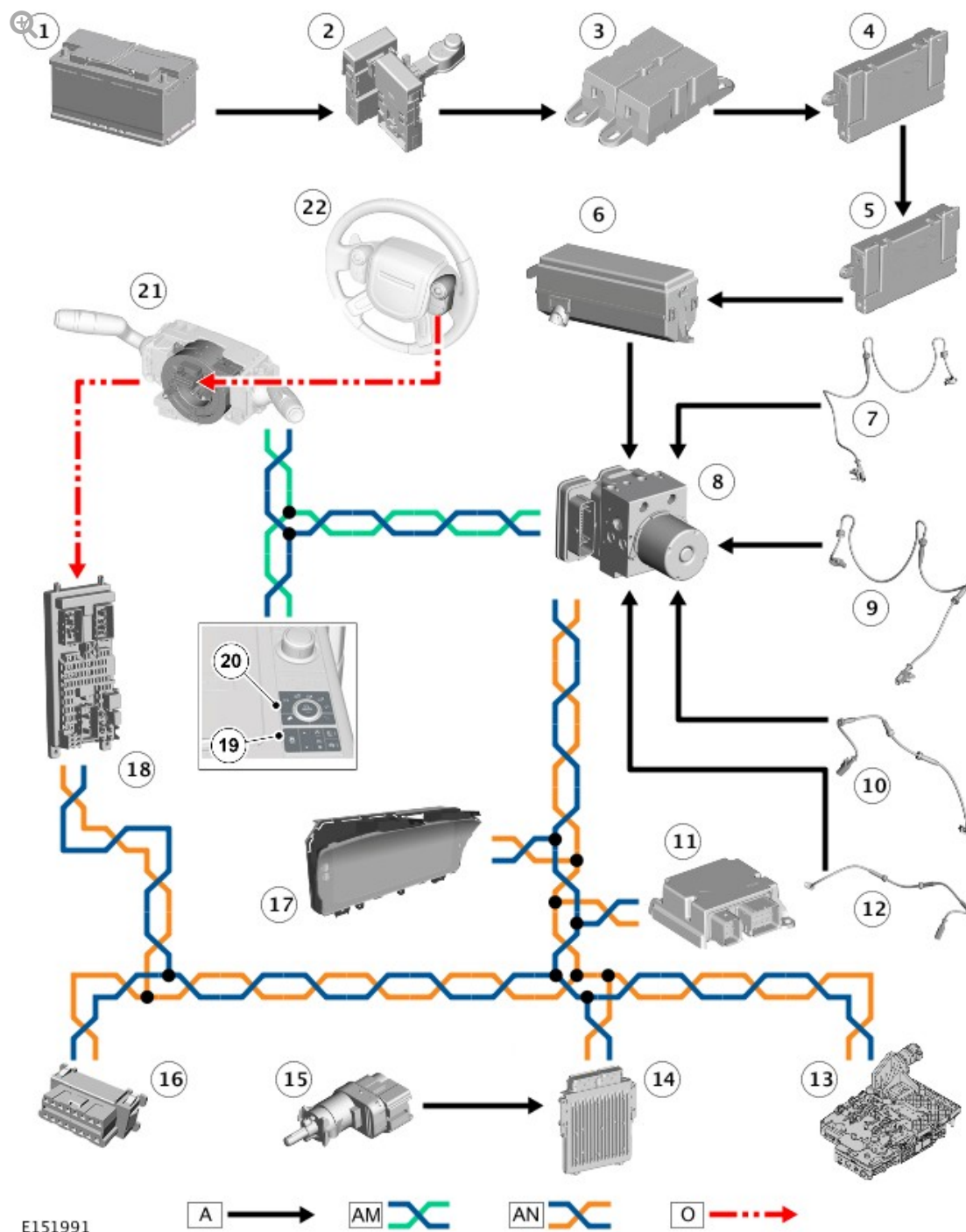
- Changing highway lanes.

- Traversing a lengthy bend.
- Acceleration.
- Braking.
- Incorrectly laden trailers.

The capability of trailer stability assist to respond early to the beginning of trailer-sway makes the system almost unnoticeable under normal driving conditions and keeps the vehicle and trailer under safe control. Trailer stability assist requires no input from the driver and operates up to the maximum vehicle speed.

Trailer stability assist will not operate while DSC is switched off.

CONTROL DIAGRAM



A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS; AM = HIGH SPEED CAN CHASSIS SYSTEMS; O = LIN (LOCAL INTERCONNECT NETWORK) BUS.

ITEM	DESCRIPTION
1	Battery
2	Megafuse
3	Battery Junction Box 2 (BJB2)

4	Battery Junction Box (BJB)
5	Auxiliary Junction Box (AJB)
6	Engine Junction Box (EJB)
7	Wheel speed sensor
8	Anti-lock brake system (ABS) control module
9	Wheel speed sensor
10	Wheel speed sensor
11	Restraints Control Module (RCM)
12	Wheel speed sensor
13	Transmission Control Module (TCM)
14	Engine Control Module (ECM)
15	Brake pedal switch
16	Diagnostic connector
17	Instrument Cluster (IC)
18	Central Junction Box (CJB)
19	Dynamic Stability Control (DSC) switch
20	Hill descent control switch
21	Steering Angle Sensor Module (SASM)
22	Speed control +/- switches